

## **Abstract**

This project investigates the factors affecting movie revenue, including production budget, cast popularity, ratings, and release season. The aim is to identify which factor affects revenue the most, compare the movie revenue achieved by each release season, and detect patterns and agreements across factors. A dataset collected from Box Office Mojo, IMDb, and TMDb of 5400 records and 13 features has been used as the primary source of data. Exploratory data analysis, correlation analysis, feature engineering, and predictive models have been applied and used to reach the desired results. The results show that production budget was the strongest predictor of revenue while ratings had a weak effect on revenue. Release seasons had a small, noticeable effect on movie revenue, with summer movies performing the best overall. Critics and audiences tended to have an agreement on movie ratings. Overall, movie success is mainly driven by financial investment along with some other small affecting factors.

**Keywords:** Movie revenue prediction; machine learning; production budget; critics' ratings; audience ratings; IMDb ratings; release season; feature engineering

## **1. Introduction**

There are many factors affecting a movie's performance. These factors can include production budget, cast popularity, ratings, and release season. For example, the movie 'Project Hail Mary' gained over 500 million dollars after having a production budget of around 200 million dollars, while other movies failed due to low production budgets. Accordingly, this project will analyze the correlation between each factor and box office success and detect patterns across these factors.

## **2. Problem Statement**

However, some factors have a larger effect on the movie's success. As a result, it remains unclear how significant the impact of each factor is on box office success. Additionally, some factors include distinct categories with different patterns that affect box office success, such as audience ratings and critics' ratings. Therefore, this project will compare the effect of each factor on box office success across genres.

## **3. Research Questions**

- What factors most strongly predict box office success — budget, cast popularity, or IMDb ratings?
- Do movies released in summer perform better financially than those in winter?
- Is there a significant difference between critics' and audience ratings across genres?

#### **4. Dataset Overview**

- Dataset name: Movies Success Factors Dataset.
- Sources: Box Office Mojo, IMDb, and TMDb.
- Number of records: 5400 records.
- Number of features: 13 features.
- Description: Each row contains the movie title, year it was released in, audience rating obtained from IMDb, critics' rating, season it was released in, budget, genres, an engineered cast popularity score, an indicator for whether the movie had a critics' rating or not, log-scaled revenue and budget, and a revenue class of high or low.
- Key variables: Revenue, ratings, season, budget, cast popularity, and revenue class.

#### **5. Data Collection Method**

Most of the data has been scraped from Box Office Mojo and IMDb's websites. Since most of the pages on these websites use dynamic tables, the data was scraped through a locally downloaded version of the websites. However, Box Office Mojo's website did not use dynamic tables for the season of each movie. This allowed the direct scraping of the website to collect the season's data for each movie. For the remaining data, a Kaggle dataset was used and merged.

#### **6. Sampling Criteria**

Due to time, computational efficiency, and bias constraints, a time-based filtering for the selection of movies has been applied. Only movies released in the 2000s were included in the dataset.

- Time constraints: It is extremely difficult to scrape the data for each movie ever released in a short period of time.
- Computational efficiency constraints: Due to the alternative approach of downloading a local version of the website, it is extremely difficult to download each page of each year on the websites of Box Office Mojo and IMDb.
- Bias constraints: Due to inflation, it is unfair to compare the revenues of movies released before and after the 2000s.

## **7. Data Preprocessing**

### **7.1 Null Value Handling**

- Genres: Each missing genre value was imputed with the mode.
- Audience/IMDb rating, cast popularity score, and budget: A temporary revenue bin column was created with 5 bins. Each missing rating, cast popularity, and budget value was then imputed with the median of its bin.
- Season: A new categorical label, unknown, was created and used.
- Critics' rating: A new row has been created to indicate whether a movie is missing the critics' rating or not. Additionally, null values were imputed with the median of their revenue bin.

### **7.2 Duplicate Handling**

There were no duplicates in the dataset after the data collection process.

### **7.3 Outliers Handling**

The outliers were detected by applying the following calculation process:

- The upper and lower quartiles were identified and saved.
- The interquartile range was calculated and stored.

- The lower value was calculated by subtracting the lower quartile by 1.5 and multiplying the result by the interquartile range.
- Similarly, the upper value was calculated by adding the upper quartile by 1.5 and multiplying the result by the interquartile range.
- The outliers were then detected by finding the values that were lower than the lower value or higher than the upper value.

After detecting the outliers for the budget and revenue, they were kept as they were since it is possible that these outliers were blockbuster movies or movies with a high production budget. However, two new log-scaled rows of the revenue and budget were created to reduce their impact and compress them.

## **7.4 Data Transformation**

- Cast popularity score: The popularity score of an actor has been calculated by dividing the movie's IMDb rating by the number of actors and adding that number to the actor's score. The movie's cast popularity score is then calculated by computing the average of the cast's total popularity score.
- Seasons: The movie's release season was obtained through a function that returned the value 'winter' for each movie released between December and February, the value 'spring' for each movie released between March and May, the value 'summer' for each movie released between June and August, and the value 'fall' for each movie released between September and November.
- Revenue class: The revenue class of a movie was considered 'high' if the movie exceeded the median of all revenues and considered 'low' if the movie was below the median.

## **8. Exploratory Data Analysis**

### **8.1 Univariate Analysis**

- Revenue histogram and log revenue histogram: A histogram that included the frequency of each movie's revenue was used to find the mode of revenue and the distribution of revenue skewness. Since the data was right-skewed, it could be

inferred that most movies earn low to moderate revenue. Similarly, another histogram was also used for the log revenue to reduce the outliers' impact. Additionally, this served as a visualization method for the outliers' influence.

- Number of movies per season bar graph: A bar graph was used to find the number of movies for each season collected in the dataset. This acted as a preliminary visualization before making conclusions on the revenue by season since some seasons may have more movies. All of the seasons had around the same number of movies, which ensured data balance.
- Critics' rating and audience rating difference histogram: A histogram of the difference between the critics' rating and audience rating was used to clarify how much critics and audiences disagree. The histogram proved that both the critics and audience agreed on the rating of most movies. However, in some movies there was a large difference between their ratings. Additionally, since the distribution was centered near zero, it could be inferred that there was a strong agreement overall.

## **8.2 Bivariate Analysis**

- Critics' rating and audience rating correlation heatmap: A heatmap representing the correlation between the critics' ratings and audience ratings was used to visualize the strength and direction of the relationship between them. Since the correlation coefficient was 0.64, there was a strong agreement between the critics and the audiences for most of the movie's ratings.
- Budget and revenue scatter plot: A scatter plot of all the movies' budgets and revenues was used to visualize the relationship between the production budget and revenue. Since the scatter points tended to move upwards, it was inferred that movies with larger production budgets had more revenue. Additionally, the log-scaled revenue was used in a different scatter plot to reduce the outliers' impact.
- IMDb Rating and revenue scatter plot and regression line: A regression line plotted on a scatter plot of all the movies' IMDb ratings and revenue indicated that there was no meaningful linear relationship between IMDb ratings and revenue due to a barely positive slope and a noisy scatter plot. Therefore, it could be concluded that the movie rating has no effect on its revenue.

- Cast popularity and revenue scatter plot and regression line: Similarly, it was inferred that the cast popularity has no influence on the movie's revenue due to a noisy scatter plot and weak linear relationship.
- Average revenue by season bar graph: A bar graph of the average revenue by each season was used to avoid the unequal group bias of a total revenue bar graph and visualize which season had better-performing movies. The graph showed that summer movies had the highest average revenue, indicating that movies released in the summer usually perform better than those released in any other season.
- Season and revenue box plot: A box plot of the revenue by each season was used to compare the statistics of each season's revenue, such as the season's median revenue, quartiles, and outliers. Through the visualization of the graph, it could be inferred that summer movies have the most consistent revenues. Additionally, summer movies had higher median revenue and a larger upper quantile than winter movies, implying that summer movies perform better. However, winter movies had larger outliers than summer movies, with a movie that performed better than any other summer movie. Moreover, a log-scaled version of the plot was created to avoid the outliers' effect.
- Season and revenue class bar graph: A bar graph of each season's count of high-revenue and low-revenue movies was used to compare success frequency across seasons. Since winter had the most movies in the dataset, more winter movies succeeded than any other seasons' movies, including summer movies. However, the ratio between movie success and failure was higher in summer movies than in winter movies; summer movies had way fewer failures than winter movies.

### **8.3 Multivariate Analysis**

- Revenue, budget, rating, and cast popularity correlation heatmap: A heatmap representing the correlation between the movies' budget, rating, cast popularity, and revenue was used to identify which factors directly affect revenue the most. Through the analysis, it was concluded that budget affects revenue the most due to the high coefficient of correlation between them.
- Revenue, budget, rating, and cast popularity scatterplot matrix: A scatterplot matrix of multiple variables was used to show their relationship with each other. The scatterplot matrix proved that the budget was the highest affecting factor to the movie's revenue because of their positive relationship and slope.

## **9. Machine Learning Model**

### **9.1 Model Selection**

- Random forest model: A random forest model was used as a feature importance tool to rank which factor affects revenue the most: budget, cast popularity, or IMDb rating. This specific model was selected for this task because it automatically calculates which variables contribute the most to predictions. Additionally, it handles nonlinear relationships, which is a requirement for this task.
- Classification model: Since the target feature is categorical, a classification model was selected to predict the revenue class of a movie based on its release season.
- K-Means clustering model: To discover patterns for unlabeled data, a clustering model is usually used. Therefore, a k-means clustering model was used to identify rating patterns between critics and audiences.

### **9.2 Model Training**

Each model was trained using a train-test split method: 80% of the dataset was used to train the model, and 20% of the dataset was used to test the model. However, the classification model needed an initial setup; the seasons column had to be split into 5 columns, including each season as a unique column and an 'unknown' season for the movies with a null season value. Additionally, the first season column was ignored as a method to prevent multicollinearity and optimize the model.

### **9.3 Model Evaluation**

- Random forest model: An  $R^2$  score was used to measure the proportion of variance in the target variable. The  $R^2$  score indicated that approximately 60% of movie revenue variance is explained by the budget, cast popularity, or IMDb rating.
- Classification model: An accuracy score was used to evaluate the classification model's accuracy. The accuracy score measured the proportion of correctly predicted movies. The model achieved an accuracy score of approximately 75%, indicating that around 75% of the movies were correctly predicted.
- K-Means clustering model: For clustering, a visual approach was used for evaluation. A scatter plot was used to inspect whether clusters were near or

separated. Since the clusters were close to each other, it could be inferred that the critics' and audiences' ratings were closely related.

## **10. Results**

- Budget was the strongest indicator of revenue, with a coefficient of correlation of 0.71.
- Ratings have a weak effect on movie revenue. Although some highly rated movies have high revenues, IMDb ratings and movie revenue had a coefficient of correlation of 0.19.
- Seasons have a mild but noticeable effect on movie revenue; summer movies have higher average revenue than other seasons. Additionally, summer movies have higher median revenue and more consistent performances. In spite of having the greatest number of high-revenue movies, winter had more variability and failures than any other season.
- Critics and audiences tend to agree on the rating of most movies, with a coefficient of correlation of 0.64.

## **11. Conclusion and Recommendations**

Throughout this project, the correlation between each factor and movie revenue has been evaluated, patterns have been detected, and several findings have been identified. Budget had the strongest relationship with movie revenue, making it the best predictor of revenue. Conversely, ratings had a weak impact on movie revenue, and seasons had a small, considerable effect. Movies released in the summer were proven to have performed better than those released in winter; summer movies had a higher average revenue and more consistent performances. Critics and audiences agreed on the ratings of most movies, with some extremely rare cases of disagreement; a strong positive relationship between critics' ratings and audiences' ratings was identified. Therefore, it could be concluded that movie success is affected by multiple factors but dominated by production budget. As a result, it is recommended that movie producers should prioritize movie budget over other revenue-affecting factors. Additionally, marketing strategies should be focused on visibility rather than aiming for higher ratings since ratings have a weak effect on movie revenue.